Bond Valuation and Interest Rate Sensitivity Analysis

Submitted to:

Ma'am Humaira Kousar

Submitted By:

Names	Registration No.
Alizeh Imran	FA24-BBD-021
Jeremiah Huang	FA24-BBD-075
Muhammad Hamza	FA24-BBD-128
Moaz Ali Tamour	FA24-BBD-099

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Executive Summary

This report examines the valuation of three debt securities currently traded on Pakistan Stock Exchange (PSX) website. The bonds were selected from three different categories: GOP Ijarah Sukuk, Public Debt Securities and Private Debt securities. These are the Government of Pakistan Fixed Rental Rate Sukuk (**P03FRR100527**), the K-Electric Ltd Sukuk (**KELSC6**), and the Bank Alfalah Term Finance Certificate (**BAFLTFC6**). In which **P03FRR100527 and BAFLTFC6** have a semiannual compounding frequency and **KELSC6** has a quarterly compounding frequency. It evaluates these securities against a benchmark **KIBOR of 11.24%** to determine their intrinsic values and sensitivity to market volatility as of March 6, 2026.

This report is divided into 5 major sections, first in which we had to fill up the relevant details for each of the bonds, then we evaluated the intrinsic values of each bond (as of March 6, 2026) and critically examined the price behavior of bonds against different periods of maturity which explained whether a bond sells at a premium/discount based on its coupon rate being higher/lower than the market rate, then we studied how the market rate of return has an inverse relationship with the price of the bond, after that we discovered how bonds with longer maturity periods are more affected by changes in interest rates and lastly how bonds are assigned credit ratings by credit rating agencies.

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1. Selection of Bonds

Bond Name	**P03FRR100527**	**BAFLTFC6**	**KELSC6**
Issuer	Government of Pakistan	Bank Alfalah Limited	K-Electric Limited
Type	GOP sukuk	Public debt	Private debt
Face Value	PKR 5,000	PKR 5,000	PKR 75,000
Coupon Rate	15.8500%	11.0700%	10.5600%
Issue Date	10-May-2024	26-Mar-2018	23-Nov-2022
Maturity	10-May-2027	26-Mar-2043	23-Nov-2029
Credit Rating	Sovereign	AA+	AA−
Rating Agency	Government of Pakistan	PACRA	PACRA

Market rate of return (k_d):

For the evaluation of non-zero coupon bonds, we are required to have a market rate of return which had not been explicitly stated on the PSX website. Therefore, we have used KIBOR (Karachi Interbank Offer Rates) as the discount rate for bonds evaluation. This is because The State Bank of Pakistan has made KIBOR the reference rate for corporate lending and most TFCs and Sukuks are priced relative to KIBOR.

KIBOR

6-Mar-2026

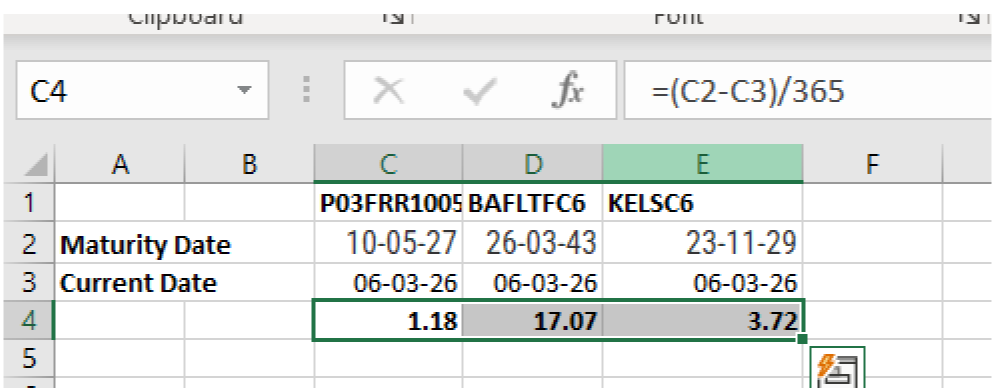
Tenor	Bid	Offer
1 - Week	10.36	10.86
2 - Week	10.36	10.86
1 - Month	10.38	10.88
3 - Month	10.48	10.73
6 - Month	10.55	10.80
9 - Month	10.71	11.21
1 - Year	10.74	11.74

Data Source: Refinitiv

All three of our debt securities had a maturity of greater than a year. So, we selected the last row from this list. In financial markets, a "Bid" is (what a bank pays to borrow) and an "Offer" is (what a bank charges to lend). The market rate of return was evaluated as the average of the two values giving 11.24%

Maturity to years (n):

In order to calculate the intrinsic value (price of the bind today) we noted down the maturity date of each bond and subtracted the current date (06-Mar-2026) from it and then divided the result by 365 to get the exact value of n in years.



C4 =(C2-C3)/365

	A	B	C	D	E	F
1			P03FRR1005	BAFLTFC6	KELSC6	
2	Maturity Date		10-05-27	26-03-43	23-11-29	
3	Current Date		06-03-26	06-03-26	06-03-26	
4			1.18	17.07	3.72	
5						

2. Bond Valuation

(a) Intrinsic Value

(i) 3 Year Fixed Rental Rate - Mat.10-May-27

6					
7		Bond Valuation			
8					P03FRR100527
9	Face Value				5000
10	Coupon rate				15.85%
11	Coupon PMT				792.50
12	Years to Maturity				1.18
13	Frequency				semi
14	Market yield (KIBOR RATE)				
15			Bid	10.74%	
16			Offer	11.74%	
17					11.24%
18		Intrinsic Value			$5,248.26
19					
20					

(ii) Bank Al-Falah (TFC6

Bank Al-Falah(TFC6)			
Maturity Date			26-Mar-43
Current Date			6-Mar-26
No of years (n)			17.07
Face Value			5,000
Coupon rate			11.07%
Coupon PMT			553.5
Market Yield			
Years to Maturity			17.07
Frequency			Semianually
Market rate(r KIBOR rate)			
Bid			10.74%
Offer			11.74%
R			11.24%
Intinsic value			$4,936.08

(iii) K-Electric LtdSC6

K Electric	
Maturity Year	2029
Current Year	2026
Years To Maturity	3
Face Value	75,000
Coupon Rate	10.56%
Coupon Payment	7,920
Market Rate	11.24%
Frequency	Quarterly
Adjusted Coupon PMT	1,980
Adjusted Market Rate	2.81%
Number of Periods	12
Intrinsic Value	73,715

(b) Behavior of Bond Price over Time:

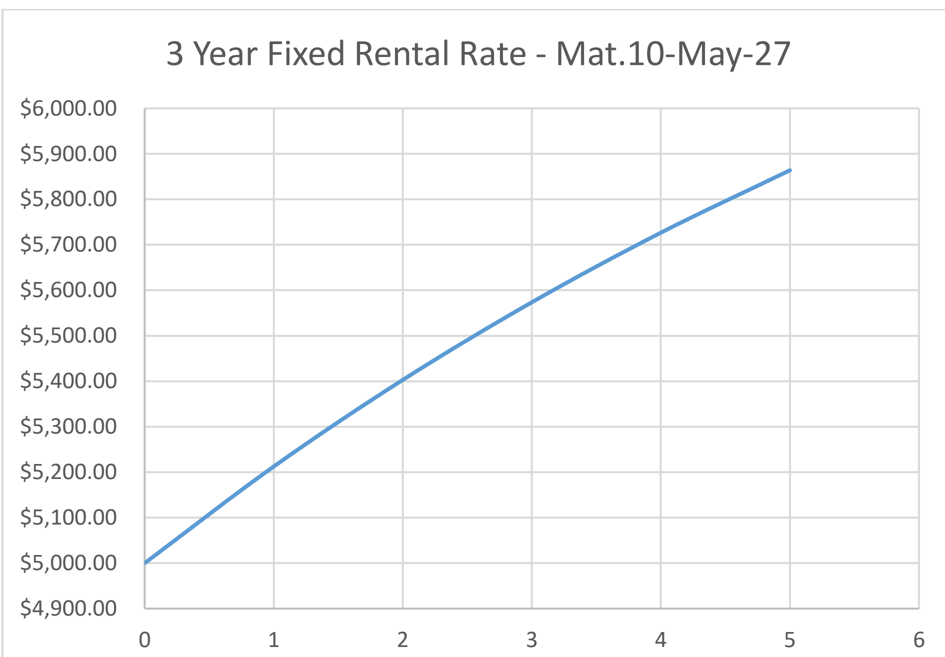
The behavior of all bond prices reflects the “Pull to Par” effect. This refers to whether a bond sells at a discount or premium, it’s price will converge to the face(par) value on the day of maturity(t=0).

(i) 3 Year Fixed Rental Rate - Mat.10-May-27

Since the coupon rate (15.85%) is greater than the market rate of return (11.24%). So, the price the of the bond (Rs. 5,248.26) would be greater than par value (Rs. 5,000). Therefore, the bond sells at a premium

3 Year Fixed Rental Rate - Mat.10-May-27

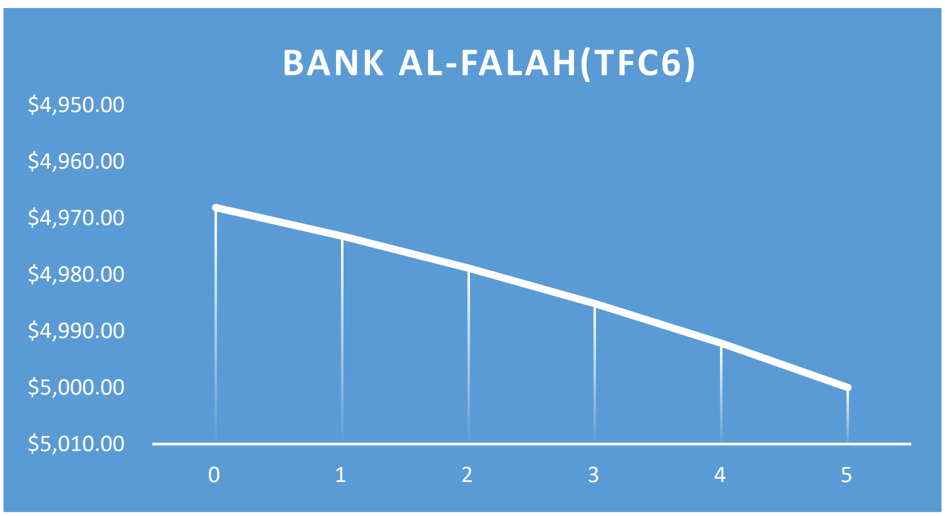
Compounding Frequency		semi		
Years to Maturity	Face Value	Coupon PMT	Market Rate of Return	Price
5	5000	792.5	11.24%	$5,863.73
4	5000	792.5	11.24%	$5,726.57
3	5000	792.5	11.24%	$5,573.55
2	5000	792.5	11.24%	$5,402.85
1	5000	792.5	11.24%	$5,212.43
0	5000	792.5	11.24%	$5,000.00



The graph shows that the price goes up from Rs. 5,000 down to Rs. 5,863.73. Indicating a rise as it moves from time 0 to 5. However, as time reaches maturity (t=5→0), the price is actually falling down to Rs. 5,000. A price of Rs. 5,863.73 indicates that the bond sells at a premium, as indicated by the line sloping downwards to the face value.

(ii) Bank Al-Falah (TFC6

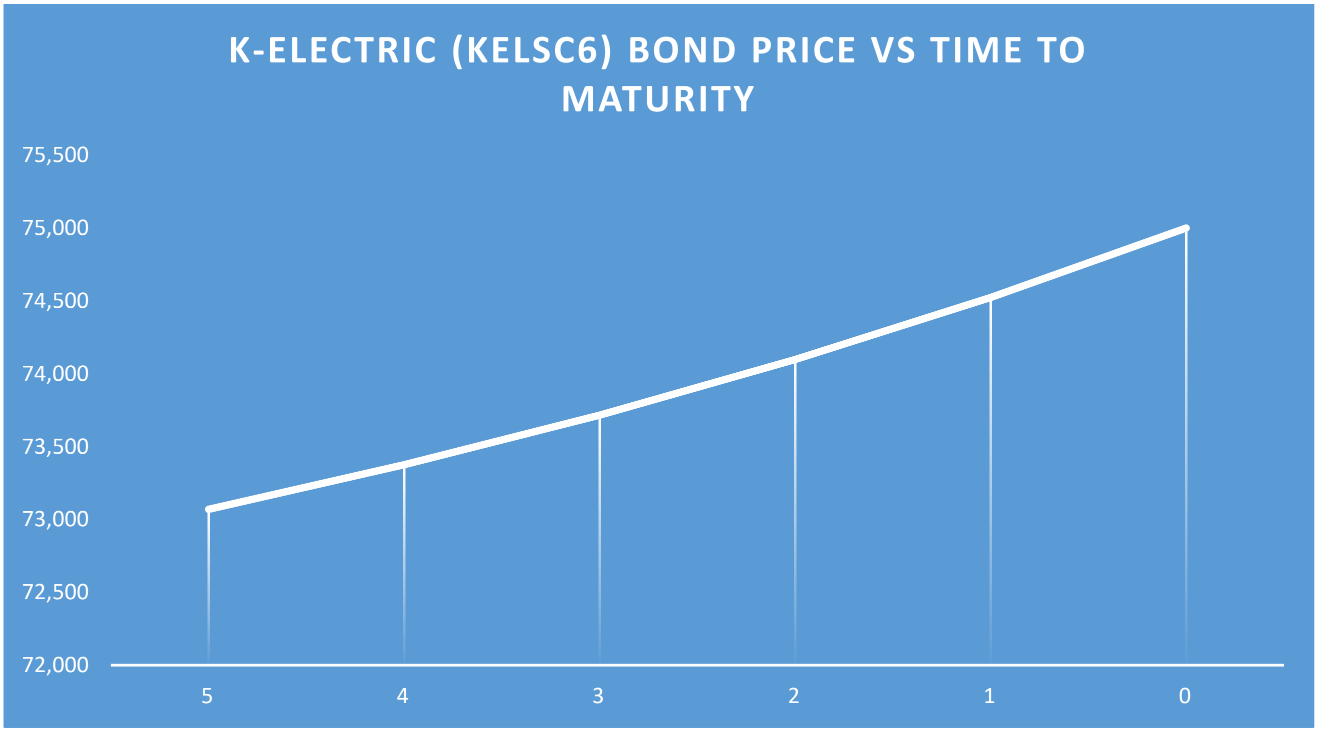
Bank Al-Falah(TFC6)						
Compounding frequency			Semi			
Years to Maturity		Face Value	Coupon PMT	Market Rate of return		PV
5		5000.00	553.50	11.24%		$4,968.15
4		5000.00	553.50	11.24%		$4,973.21
3		5000.00	553.50	11.24%		$4,978.85
2		5000.00	553.50	11.24%		$4,985.14
1		5000.00	553.50	11.24%		$4,992.17
0		5000.00	553.50	11.24%		$5,000.00



The graph shows that as the **time to maturity decreases from 5 to 0**; the **bond price gradually increases toward its face value of Rs. 5,000**. This happens because the bond is selling at a **discount** due to the **coupon rate (11.07%) being slightly lower than the market rate (11.24%)**.

(iii) K-Electric LtdSC6

Years to Maturity	Number of Periods	Face Value	Intrinsic Value	Coupon PMT	Adjusted Coupon PMT	Market Rate	Adjusted Market Rate	PV
5	20	75000	73,716	7920	1980	11.24%	2.81%	73,069
4	16	75000	73,716	7920	1980	11.24%	2.81%	73,375
3	12	75000	73,716	7920	1980	11.24%	2.81%	73,716
2	8	75000	73,716	7920	1980	11.24%	2.81%	74,098
1	4	75000	73,716	7920	1980	11.24%	2.81%	74,524
0	0	75000	73,716	7920	1980	11.24%	2.81%	75,000



The time to maturity analysis shows how the bond price changes as the bond approaches its maturity date. The table and graph illustrate that as the number of years to maturity decreases from 5 years to 0, the price of the bond gradually increases toward its face value of PKR 75,000. This occurs because the issuer is obligated to repay the full face value of the bond at maturity. Since the bond is currently selling at a discount due to the coupon rate being slightly lower than the market rate, the bond price slowly converges toward the face value as the maturity date approaches. This behavior demonstrates a common characteristic of bonds where prices move closer to their face value as maturity approaches.

3. Interest Rate Impact Analysis

(a) Intrinsic Value Under Different Market Rates

The base market rate is 11.24%. In order to show the behavior of price in accordance with different market rates we are using the following rates: 9.24%, 10.24, 11.24% (base), 12.24%, and 13.24%.

This section demonstrates the effect of different market rates on the price of the bond, proving that the market rate of return has an inverse relationship to the bond price.

(i) 3 Year Fixed Rental Rate - Mat.10-May-27

	Interest Rate Impact Analysis				
	3 Year Fixed Rental Rate - Mat.10-May-27				
	Compounding Frequency		semi	Coupon Ra 15.85%	
	Years to Maturity	Face Value	Coupon PMT	Market Rate of Return	PV
	1.18	5000	792.5	9.24%	$5,361.63
	1.18	5000	792.5	10.24%	$5,304.50
Base	1.18	5000	792.5	11.24%	$5,248.26
	1.18	5000	792.5	12.24%	$5,192.89
	1.18	5000	792.5	13.24%	$5,138.38

(ii) Bank Al-Falah (TFC6

	Bank Al-Falah(TFC6)				
	Compounding frequency		Semi	Coupon rate 11.07%	
	Years to Maturity	Face Value	Coupon PMT	Market Rate of return	PV
	17.07	5000.00	553.50	9.24%	$5,778.29
	17.07	5000.00	553.50	10.24%	$5,331.58
base	17.07	5000.00	553.50	11.24%	$4,936.07
	17.07	5000.00	553.50	12.24%	$4,584.96
	17.07	5000.00	553.50	13.24%	$4,272.37

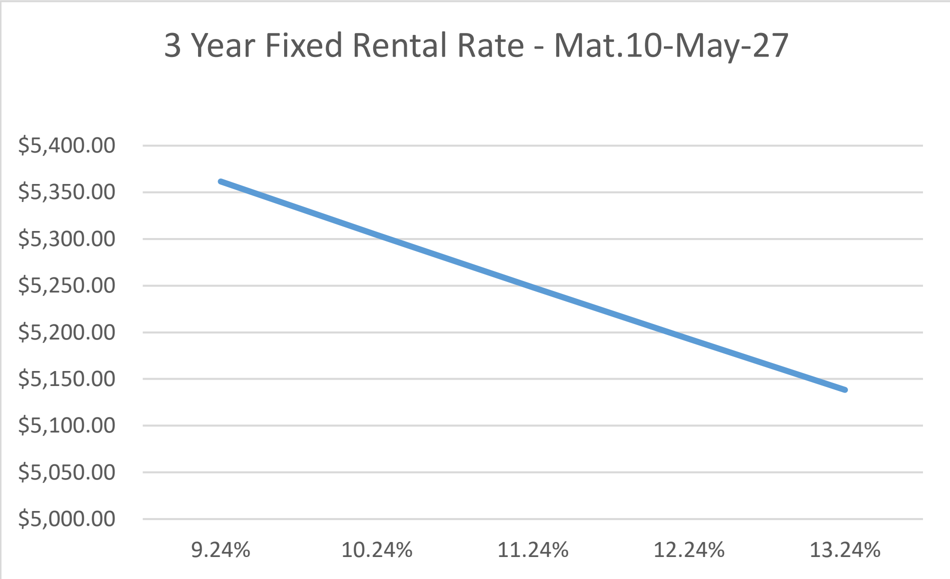
(iii) K-Electric LtdSC6

Maturity Years	Intrinsic Value	Market Rate	Adjusted Market Rate	Face Value	Coupon PMT	Adjusted Coupon PMT	PV
3.72	73,716	9.24%	2.31%	75,000	7,920	1,980	78,087
3.72	73,716	10.24%	2.56%	75,000	7,920	1,980	75,735
3.72	73,716	11.24%	2.81%	75,000	7,920	1,980	73,467
3.72	73,716	12.24%	3.06%	75,000	7,920	1,980	71,280
3.72	73,716	13.24%	3.31%	75,000	7,920	1,980	69,170

(b) Analysis of Price movement on Graph

(i) 3 Year Fixed Rental Rate - Mat.10-May-27

The coupon rate (15.85%) is significantly higher than the range of market rates, so the bond remains at a premium throughout the different market rate. All of the prices are greater than the par value (Rs. 5,000). The graph shows a downward slope which indicates that an increase market rate results in lower prices.

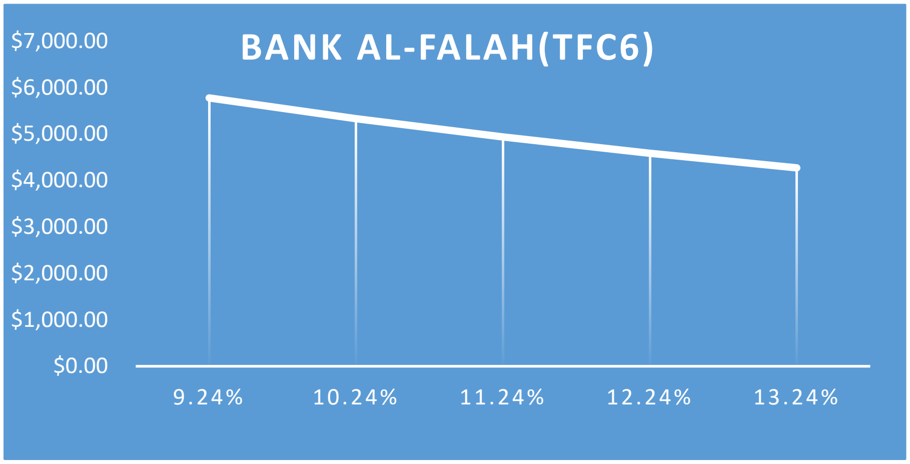


(ii) Bank Al-Falah (TFC6

The graph represents that if the

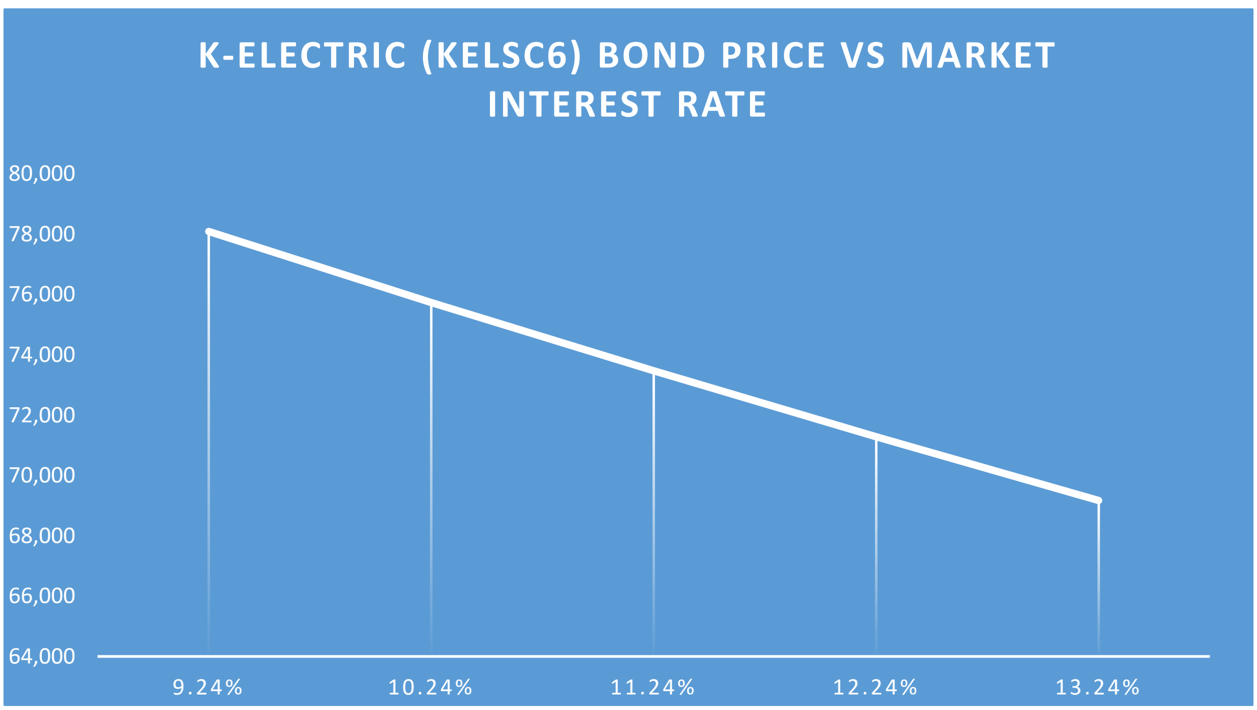
- market rate is at **9.24%** (lower than the coupon rate 11.07%) the value of the bond will be **($5778.29)** which is higher than the par value **($5000)** this represents the bond will sell at premium
- market rate at **10.24%** it will still sell at premium but with lower bond value of (**$5331.58**) as compared to previous rate
- market rate at 11.24% (higher than the coupon rate) now the bond value (**$4936.07**) is lower than the par value of the bond so it will sell at discount
- market rate at 12.24% it will still sell at discount but with lower bond value $4584.96 as compared to previous rate
- market rate at 13.24% it will still sell at discount but with lower bond value $4272.37 as compared to previous rate

As the market rate of return increases from 9.24% to 13.24%, the bond's present value decreases from $5,778.29 to $4,272.37. This inverse relationship demonstrates that when market rates rise above the bond's coupon rate (11.07%), the bond trades . if the Market rate of return and coupon are equal then the bond will sell at the par value of the bond



(iii) K-Electric LtdSC6

The interest rate sensitivity analysis sees that how the changes in the market interest rates affect the price of K Electric's bond. The results show us that whenever the market interest rate is lower the price of bond becomes higher as the fixed coupon payments become more attractive to investors. And when the market interest rate increases the PV of the bond decreases. For example at market rate 9.24%, the price of bond rises to PKR 78,087 and when the market rate is 13.24%, the price of the bond decreases to PKR 69,170.

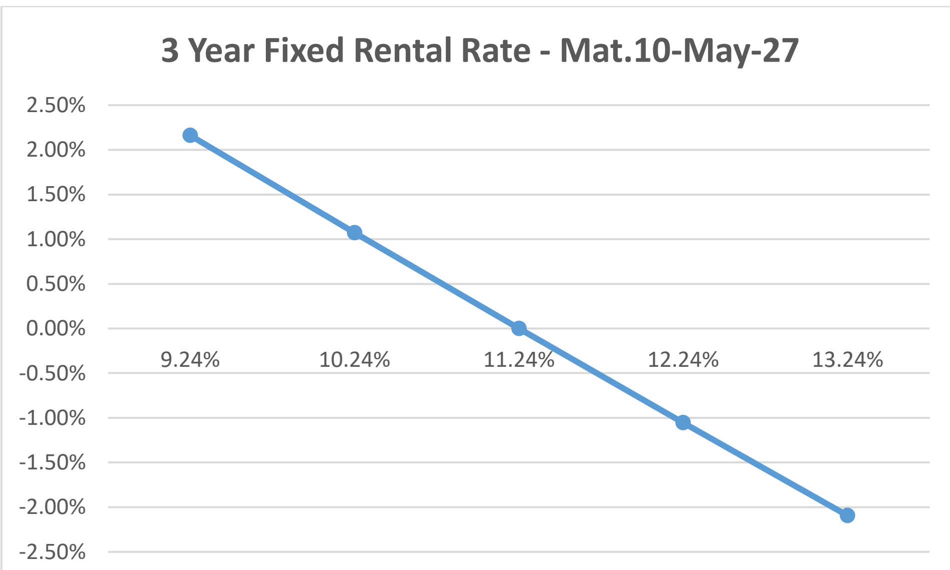


4. Maturity Comparison

(a) Maturity Affecting Price Sensitivity Using Graphs

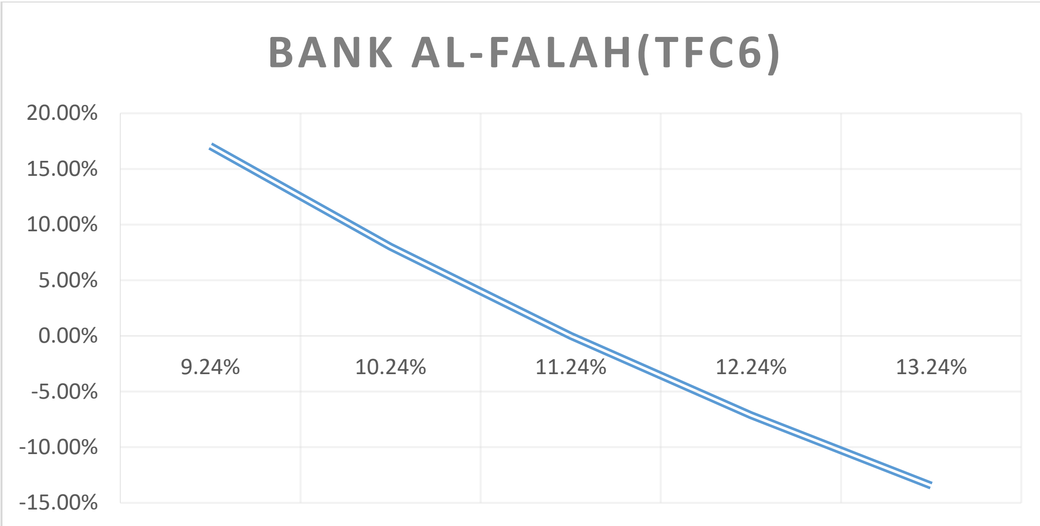
(i) 3 Year Fixed Rental Rate - Mat.10-May-27

3 Year Fixed Rental Rate - Mat.10-May-27			
Market Ra	Price	Base Price	% change
9.24%	$5,361.63	$5,248.26	2.16%
10.24%	$5,304.50	$5,248.26	1.07%
11.24%	$5,248.26	$5,248.26	0.00%
12.24%	$5,192.89	$5,248.26	-1.05%
13.24%	$5,138.38	$5,248.26	-2.09%



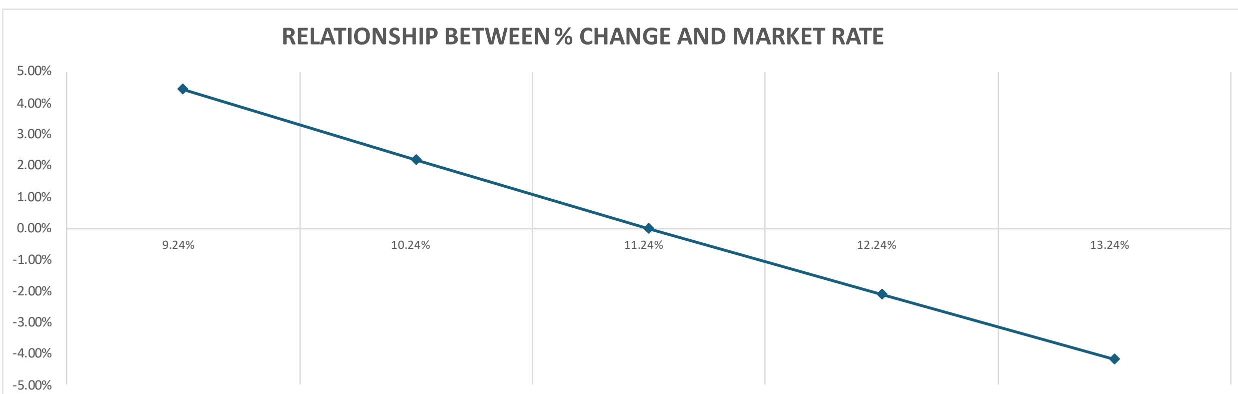
(ii) Bank Al-Falah (TFC6

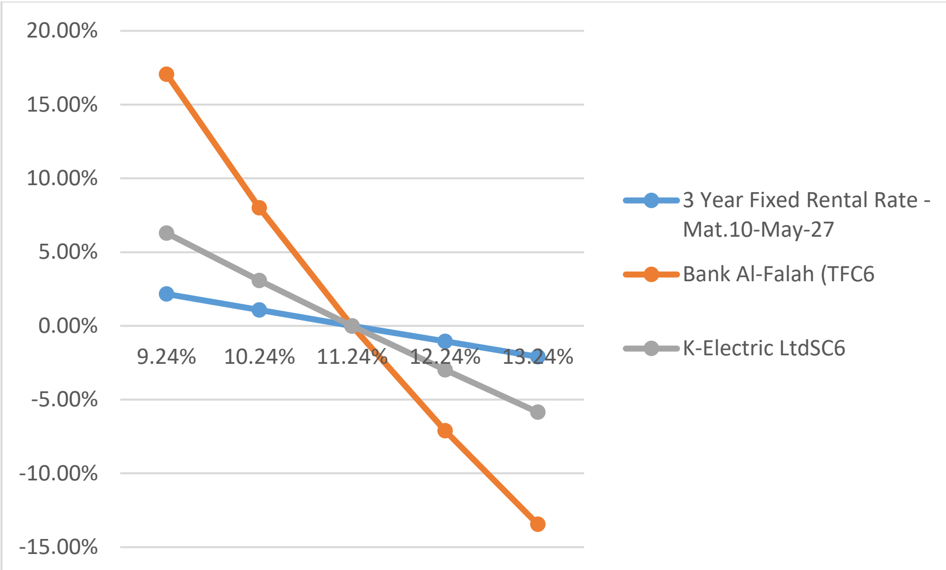
Market Rate	Price	base Price	%change
9.24%	$5,778.29	$4,936.07	17.06%
10.24%	$5,331.58	$4,936.07	8.01%
11.24%	$4,936.07	$4,936.07	0.00%
12.24%	$4,584.96	$4,936.07	-7.11%
13.24%	$4,272.37	$4,936.07	-13.45%



(iii) K-Electric LtdSC6

Intrinsic Value	Market Rate	PV	% Change
73,716	9.24%	78,087	5.93%
73,716	10.24%	75,735	2.74%
73,716	11.24%	73,467	-0.34%
73,716	12.24%	71,280	-3.31%
73,716	13.24%	69,170	-6.17%





(b) Which Bond is Most Sensitive to Interest Rate Change

The main idea is that the longer the period of the maturity the higher the price sensitivity to interest rate changes of the bond if the interest changes the cashflows such as coupon payments will be discounted to the newer market rate if the bond have longer maturity time these payments will occur over many times small change in interest rate will cause a large change in the bond price as compared to short term bond the most payments will be received soon because there will be less time for change

Visualizing Sensitivity (Graph Explanation)

The graph helps show how strongly each security reacts when the market rate increases or decreases

- The **BAFLTFC6:** line appears much steeper which means its price changes more when interest rates move
- The **KELSC6:** line shows moderate slope indicating medium sensitivity
- The **P03FRR100527:** line is almost horizontal showing very small price changes

This happens because longer maturity securities have cash flows that occur late
when interest rate changes the present value of these future payments changes more significantly

The results show that **BAFLTFC6** is the most sensitive to interest rate changes because of its long maturity of 17.07 years
A large part of its value depends on future cash flows so changes in interest rates cause larger price movements

In contrast **P03FRR100527** has a maturity of only 1.18 years so interest rate changes have a smaller effect on its price Overall, the analysis shows that longer maturity bonds have higher price volatility than short maturity bonds

5. Credit Rating Analysis

(a). Credit rating assigned to each bond

- **P03FRR100527**: is issued by the Government of Pakistan and has a AAA (Local) sovereign rating
- **BAFLTFC6**: issued by Bank Alfalah Limited has an AA+ credit rating
- **KELSC6**: issued by K-Electric Limited has an AA+ credit rating

(b). Rating agencies

The ratings for BAFLTFC6 are assigned by PACRA / VIS.
The ratings for KELSC6 are assigned by VIS / PACRA.
The P03FRR100527 Sukuk carries a AAA (Local) sovereign rating because it is issued by the Government of Pakistan.

(c). How credit rating affects required return and price

Credit ratings tell the risk level of the issuer. securities with higher ratings are safer so investors require a lower return. because the required return is lower the price of the bond becomes higher.

If the credit rating is lower investors see more risk and demand a higher required return. this higher required return reduces the present value of future cash flows which ultimately reduces the bond price.

In this comparison BAFLTFC6 and KELSC6 both have AA+ ratings, meaning both instruments have a relatively strong credit quality. The P03FRR100527 sovereign Sukuk has a AAA (Local) rating, which indicates very high credit quality and strong repayment capacity since it is backed by the government.